CORE PURE / POLAR COORDINATES

Tangents to Polar Curves

Mark scheme 8 questions · 94 marks

EXAM BOARDS Edexcel 9FM0 · AQA 7367 · CIE 9231

Questions

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	Total	**94**	

Reading the mark scheme

These rules apply throughout, unless a note says otherwise.

Marks

Every mark is worth one, so **M1 A1** on a row is two marks. Only marks coded ft follow through.

Code	Meaning
M1	Method: a correct method, applied to this question. A slip doesn't cost it if the method is clear. A formula must be used, not just quoted.
dM1	Dependent method: as M1, but only if the method mark before it was earned, unless its note names another.
A1	Accuracy: a correct answer or step. It needs the method mark it follows.
A1ft	As A1, following through from an earlier answer that the note names.
A1 AG	The last mark of a part whose answer is printed in the question: complete working, with no errors.
B1	A correct value, fact or statement. It needs no method mark.
B1ft	As B1, following through from an earlier answer that the note names.
E1	A written explanation, interpretation or conclusion. Concluding a proof by induction needs every earlier mark.

Abbreviations

ft and AG qualify a code in the Marks column. The others appear in the notes. A grey line gives the rule where the name alone doesn't.

ft	follow through Correct work from the student's own earlier wrong value earns the mark.	**AG**	answer given The answer is printed in the question, so every step to it must be shown.
cao	correct answer only No follow-through from an earlier error.	**cso**	correct solution only No error anywhere in the part, not just in its answer.
oe	or equivalent	**awrt**	answers which round to
soi	seen or implied Earned if written, or if later working shows it was used.	**isw**	ignore subsequent working A correct answer, once reached, isn't spoiled by later working.
SC	special case Credit, as the note says, for work outside the main method.	**their**	the student's own earlier value
condone	a slip that doesn't cost the mark	[ ]	working the mark doesn't need
s.f.	significant figures	**d.p.**	decimal places

Reading the scheme

- Each row is a line of working, with the marks it earns beside it. Where a row carries two marks, each note starts with its code.
- **[*n*]** ends each part with its total.
- A shaded *Alternative* is a second method for the same total. Mark an answer by the method it follows.
- Any other valid method earns the equivalent marks, unless the question names a method or says *hence*.
- A sketch earns one B1 for each feature named below its figure.

Accuracy and answers

- Unless a question says otherwise, give answers to 3 significant figures, and angles in degrees to 1 decimal place. Angles in radians take 3 significant figures.
- A final answer with more figures is accepted if it rounds to the scheme's value; one with fewer is not. An answer spoiled by rounding too early loses its accuracy mark.
- Where a question asks for an exact answer, a decimal loses the final accuracy mark. Otherwise either is accepted.
- Wrong or missing units cost nothing unless the question asks for them.
- An omitted $\mathrm{d}\theta$, $\mathrm{d}x$ or $\mathrm{d}u$ in an integral is not penalised where the variable of integration is clear from the integrand and the limits. It is penalised where it is genuinely ambiguous: a substitution part-way through, two variables in play, or limits in the wrong variable.
- A correct answer with no working earns full marks, unless the question needs the working: show that, prove, hence, a method it names, or an exact answer. A wrong answer with no working earns nothing. Calculator answers are treated the same way.
- A correct answer, once reached, isn't spoiled by later working (isw).

Value of g

- Take $g = 9.8\,\mathrm{m\,s^{-2}}$ unless a question states a value, which is then the only one accepted. Answers that use $g = 9.8$ may be given to 2 or 3 significant figures.
- Where a question doesn't state g, an answer worked with $g = 10$ throughout earns the same marks, keeping any value printed in the question. The worked solutions list the final answers with $g = 10$.

Question	Scheme	Marks	Notes
1	$x = r\cos\theta = 3\sin 2\theta\cos\theta$ $\Rightarrow \frac{\mathrm{d}x}{\mathrm{d}\theta} = \ldots$	**M1**	Writes x in terms of θ only and differentiates, by the product rule or after any rewriting. Condone sign and coefficient slips.
	$\frac{\mathrm{d}x}{\mathrm{d}\theta} = 6\cos 2\theta\cos\theta - 3\sin 2\theta\sin\theta$	**A1**	Correct derivative in any form.
	$6\cos 2\theta\cos\theta - 3\sin 2\theta\sin\theta = 0$ $\Rightarrow 6\cos\theta(3\cos^2\theta - 2) = 0$ $\Rightarrow \cos^2\theta = \ldots$	**M1**	Sets their $\frac{\mathrm{d}x}{\mathrm{d}\theta} = 0$, uses identities to reach a single trigonometric ratio of θ, and solves for it. Condone slips.
	$\cos^2\theta = \frac{2}{3}$ ($\cos\theta = 0$ gives the pole)	**A1**	Or $\sin^2\theta = \frac{1}{3}$, $\tan\theta = \frac{1}{\sqrt{2}}$ oe. $\cos\theta = 0$ rejected or not pursued; A0 if it is offered as P.
	$r = 6\sin\theta\cos\theta = 6 \times \frac{1}{\sqrt{3}} \times \sqrt{\frac{2}{3}}$	**M1**	Substitutes their θ or trigonometric ratios, with $0 < \theta < \frac{\pi}{2}$, into $r = 3\sin 2\theta$.
	$OP = 2\sqrt{2}$	**A1**	cso. Any exact equivalent, such as $\sqrt{8}$.
	Total for Question 1	**6**	

Question	Scheme	Marks	Notes
2	$y = r\sin\theta = 2\sin 3\theta \sin\theta$ $\frac{\mathrm{d}y}{\mathrm{d}\theta} = 6\cos 3\theta \sin\theta + 2\sin 3\theta \cos\theta$	**M1 A1**	**M1** Differentiates $y = 2\sin 3\theta \sin\theta$, or an equivalent such as $\cos 2\theta - \cos 4\theta$, with respect to θ. Condone coefficient slips. **A1** Correct $\frac{\mathrm{d}y}{\mathrm{d}\theta}$, oe.
	$3\cos 3\theta \sin\theta + \sin 3\theta \cos\theta = 0$ $\Rightarrow \sin\theta\cos\theta(16\cos^2\theta - 10) = 0$ $\Rightarrow \cos^2\theta = \ldots$	**M1**	Sets their $\frac{\mathrm{d}y}{\mathrm{d}\theta} = 0$ and uses multiple-angle identities to reach and solve an equation in $\cos^2\theta$, $\sin^2\theta$ or $\cos 2\theta$. Condone slips.
	$\cos^2\theta = \frac{5}{8}$ ($\sin\theta = 0$ gives the pole)	**A1**	Or $\sin^2\theta = \frac{3}{8}$ or $\cos 2\theta = \frac{1}{4}$, with $\theta = 0$ not taken as P.
	$\sin\theta = \frac{\sqrt{6}}{4}$ $\Rightarrow r = 2(3\sin\theta - 4\sin^3\theta) = \ldots$	**M1**	Substitutes their θ, from a valid $\cos^2\theta$ or equivalent, into $r = 2\sin 3\theta$, by any means.
	$OP = \frac{3\sqrt{6}}{4}$	**A1**	Exact, oe such as $\sqrt{\frac{27}{8}}$.
	Total for Question 2	**6**	

Question	Scheme	Marks	Notes
3(a)	Area $= \frac{1}{2}\int_{-\pi}^{\pi} 4(1+\cos\theta)^2\,\mathrm{d}\theta$	**B1**	Correct integral with correct limits, oe such as $2\times\frac{1}{2}\int_0^{\pi}$. Needs the $\frac{1}{2}$.
	$= 2\int_{-\pi}^{\pi}\left(\frac{3}{2}+2\cos\theta+\frac{1}{2}\cos 2\theta\right)\mathrm{d}\theta$ $= 2\left[\frac{3}{2}\theta+2\sin\theta+\frac{1}{4}\sin 2\theta\right]_{-\pi}^{\pi}$	**M1**	Expands, uses a double-angle identity for $\cos^2\theta$ and integrates to the form $p\theta+q\sin\theta+s\sin 2\theta$. Condone sign and coefficient slips.
	$= 2\left(\frac{3}{2}\pi-\left(-\frac{3}{2}\pi\right)\right)$	**dM1**	Substitutes their limits into their integrated expression.
	$= 6\pi$	**A1**	6π, or awrt 18.8.
		[4]	
3(b)(i)	$2(1+\cos\theta)=\sin\theta$ $\Rightarrow 4(1+\cos\theta)^2 = 1-\cos^2\theta$	**M1**	Equates the two expressions for r and forms an equation in one trigonometric function, e.g. by squaring and using $\sin^2\theta = 1-\cos^2\theta$.
	$5\cos^2\theta+8\cos\theta+3=0$	**A1**	oe, terms collected.
	$(5\cos\theta+3)(\cos\theta+1)=0$ $\Rightarrow \cos\theta=-\frac{3}{5}$ ($\cos\theta=-1$ gives O)	**dM1**	Solves their three-term quadratic and rejects $\cos\theta=-1$, soi by using only the other root.
	$\sin\theta=\frac{4}{5}\Rightarrow A\left(\frac{4}{5},\ \arccos\left(-\frac{3}{5}\right)\right)$	**A1**	$r=\frac{4}{5}$ and $\theta=\arccos\left(-\frac{3}{5}\right)$, awrt 2.21 or awrt 126.9°.
		[4]	
3(b)(ii)	$x_A=\frac{4}{5}\times\left(-\frac{3}{5}\right)=-\frac{12}{25}$	**B1**	oe, such as -0.48.
	$2\cos\theta(1+\cos\theta)=$ "$-\frac{12}{25}$"	**M1**	Sets $r\cos\theta$ for C_1 equal to their x-coordinate of A.
	$25\cos^2\theta+25\cos\theta+6=0$	**A1**	oe, terms collected.
	$(5\cos\theta+3)(5\cos\theta+2)=0\Rightarrow\cos\theta=\ldots$	**dM1**	Solves their three-term quadratic in $\cos\theta$.
	$\cos\theta=-\frac{2}{5}$ ($-\frac{3}{5}$ is A)	**A1**	Rejection of $-\frac{3}{5}$ soi by using only $-\frac{2}{5}$.
	$OB=2\left(1-\frac{2}{5}\right)=\frac{6}{5}$	**A1 AG**	$\cos\theta=-\frac{2}{5}$ substituted correctly into $r=2(1+\cos\theta)$, reaching $\frac{6}{5}$ without error.
		[6]	
3(b)(iii)	$\sin\theta_B=\frac{\sqrt{21}}{5}$ $y_A=\frac{4}{5}\times\frac{4}{5},\quad y_B=\frac{6}{5}\times\frac{\sqrt{21}}{5}$	**M1**	Uses $y=r\sin\theta$ for both A and B, with $\sin\theta$ at B found from $\cos\theta=-\frac{2}{5}$. Need not be simplified.

Question 3 continued

Question	Scheme	Marks	Notes
	$y_A = \frac{16}{25}, \quad y_B = \frac{6\sqrt{21}}{25}$	**A1**	oe, or awrt 0.640 and 1.10.
	$AB = \frac{6\sqrt{21}-16}{25} = 0.460$	**A1**	awrt 0.460.
		[3]	
	Total for Question 3	**17**	

Question	Scheme	Marks	Notes
4(a)	$\frac{1}{2}\int_0^{2\pi} a^2(1+\sin\theta)^2\,\mathrm{d}\theta$	**B1**	Limits spanning one full turn, e.g. $-\frac{\pi}{2}$ to $\frac{3\pi}{2}$, accepted. Needs the $\frac{1}{2}$.
	$\frac{a^2}{2}\int\left(\frac{3}{2}+2\sin\theta-\frac{1}{2}\cos 2\theta\right)\mathrm{d}\theta$ $=\frac{a^2}{2}\left[\frac{3}{2}\theta-2\cos\theta-\frac{1}{4}\sin 2\theta\right]$	**M1**	Expands, uses a double-angle identity for $\sin^2\theta$ and integrates to the form $p\theta+q\cos\theta+s\sin 2\theta$. Condone sign and coefficient slips.
	$\frac{a^2}{2}\left[(3\pi-2)-(-2)\right]$	**dM1**	Substitutes their limits into their integrated expression and subtracts.
	$\frac{3\pi a^2}{2}$	**A1**	oe.
		[4]	
4(b)	$y=a(1+\sin\theta)\sin\theta$ $\frac{\mathrm{d}y}{\mathrm{d}\theta}=a\cos\theta\sin\theta$ $\quad+a(1+\sin\theta)\cos\theta$	**M1**	Differentiates $r\sin\theta$ with their r, by the product rule or after expanding. Condone sign slips.
	$\frac{\mathrm{d}y}{\mathrm{d}\theta}=a\cos\theta(1+2\sin\theta)$	**A1**	oe, e.g. $a(\cos\theta+\sin 2\theta)$.
	$\cos\theta=0$ or $\sin\theta=-\frac{1}{2}$ $\Rightarrow\theta=\ldots$	**M1**	Sets their $\frac{\mathrm{d}y}{\mathrm{d}\theta}=0$ and solves at least one factor for θ.
	$\theta=\frac{\pi}{2},\ \frac{7\pi}{6},\ \frac{11\pi}{6}$ ($\frac{3\pi}{2}$ rejected: $r=0$)	**A1**	These three only, or awrt 1.57, 3.67, 5.76. A0 if $\frac{3\pi}{2}$ is included.
	$r=a\left(1+\sin\frac{7\pi}{6}\right)=\ldots$	**M1**	Substitutes a value with $\sin\theta=-\frac{1}{2}$ into $r=a(1+\sin\theta)$.
	$A\left(\frac{a}{2},\frac{7\pi}{6}\right),\ B\left(\frac{a}{2},\frac{11\pi}{6}\right)$	**A1**	Both, correctly assigned to A and B. Angles awrt 3.67 and 5.76 accepted.
		[6]	
4(c)	$y=2a\sin\frac{\pi}{2}=2a,\ y=\frac{a}{2}\sin\frac{7\pi}{6}=-\frac{a}{4}$ $WZ=2a-\left(-\frac{a}{4}\right)$	**M1**	Finds $y=r\sin\theta$ at $\theta=\frac{\pi}{2}$ and at their A or B, and finds the distance between them.
	$WZ=\frac{9a}{4}$	**A1 AG**	Both y-values correct, combined without error to the printed result.
		[2]	
4(d)	$\frac{3\sqrt{3}\,a}{2}\times\frac{9a}{4}$	**M1**	Multiplies the given WX by $\frac{9a}{4}$.

Question 4 continued

Question	Scheme	Marks	Notes
	$\frac{27\sqrt{3}\,a^2}{8}$	**A1**	oe.
		[2]	
4(e)	$\frac{27\sqrt{3}\times 8^2}{8} = 216\sqrt{3}$ $[\approx 374.1]$	**B1**	$216\sqrt{3}$ or awrt 374.
	$216\sqrt{3} - \frac{3\pi\times 8^2}{2} = 216\sqrt{3} - 96\pi$	**M1**	Subtracts their area from (a), with $a = 8$, from their rectangle area.
	≈ 72.5 (cm^2)	**A1**	awrt 72.5, or $216\sqrt{3} - 96\pi$.
		[3]	
	Total for Question 4	**17**	

Question	Scheme	Marks	Notes
5(a)	$x = 4(1-\cos\theta)\cos\theta$ $= 4(\cos\theta - \cos^2\theta) \Rightarrow \frac{\mathrm{d}x}{\mathrm{d}\theta} = \ldots$	**M1**	Forms $x = r\cos\theta$ with the given r and differentiates with respect to θ, by product rule or after expanding. Condone sign slips.
	$\frac{\mathrm{d}x}{\mathrm{d}\theta} = 4(-\sin\theta + 2\sin\theta\cos\theta)$	**A1**	oe; need not be factorised.
	$4\sin\theta(2\cos\theta - 1) = 0$ $\Rightarrow \theta = 0$ (pole, rejected) or $\cos\theta = \frac{1}{2}$	**M1**	Sets their $\frac{\mathrm{d}x}{\mathrm{d}\theta}$, in terms of $\sin\theta$ and $\cos\theta$, equal to 0, solves for θ and rejects $\theta = 0$ as the pole.
	$\theta = \frac{\pi}{3},\ r = 4\left(1 - \frac{1}{2}\right) = 2$	**A1 AG**	cso, with $r = 2$ shown by substituting $\theta = \frac{\pi}{3}$.
		[4]	
5(b)	Area $= \int_0^{\pi/3} y\,\frac{\mathrm{d}x}{\mathrm{d}\theta}\,\mathrm{d}\theta$ $= 16\int_0^{\pi/3}(1-\cos\theta)\sin^2\theta$ $\times\,(2\cos\theta - 1)\,\mathrm{d}\theta$	**M1 A1**	**M1** Uses $\int y\frac{\mathrm{d}x}{\mathrm{d}\theta}\,\mathrm{d}\theta$ with $y = r\sin\theta$, their $\frac{\mathrm{d}x}{\mathrm{d}\theta}$ from (a) and limits 0 and $\frac{\pi}{3}$. **A1** Correct integrand in θ, oe, with limits.
	$16\int_0^{\pi/3}(2\cos^4\theta - 3\cos^3\theta - \cos^2\theta$ $+\,3\cos\theta - 1)\,\mathrm{d}\theta = 16\left[\ldots\right]_0^{\pi/3}$	**M1**	Expands, uses double-angle identities for even powers and reverse chain rule or $1-\sin^2\theta$ for odd powers, and integrates. Condone sign and coefficient slips.
	$16\left[\sin^3\theta + \frac{1}{4}\sin 2\theta + \frac{1}{16}\sin 4\theta - \frac{3}{4}\theta\right]_0^{\pi/3}$	**A1**	Correct integral, oe; need not be simplified.
	$16\left[\left(\frac{\sqrt{3}}{2}\right)^3 + \frac{1}{4}\cdot\frac{\sqrt{3}}{2} - \frac{1}{16}\cdot\frac{\sqrt{3}}{2} - \frac{\pi}{4}\right] - 0$	**dM1**	Substitutes $\frac{\pi}{3}$ and 0 into their integrated expression and subtracts; the zero lower value may be implied.
	$= \frac{15\sqrt{3}}{2} - 4\pi = \frac{1}{2}(15\sqrt{3} - 8\pi)$	**A1 AG**	cso: correct exact sine values leading without error to the printed result.
		[6]	
	Total for Question 5	**10**	

Question	Scheme	Marks	Notes
6(a)	$y = (1 + 2\cos 2\theta)\sin\theta$ $\frac{\mathrm{d}y}{\mathrm{d}\theta} = -4\sin 2\theta \sin\theta + (1 + 2\cos 2\theta)\cos\theta$	**M1 A1**	**M1** Differentiates $r\sin\theta$ with $r = 1 + 2\cos 2\theta$, by the product rule or after expanding. Condone sign and coefficient slips. **A1** Any correct form.
	Sets $\frac{\mathrm{d}y}{\mathrm{d}\theta} = 0$: $3\cos\theta(1 - 4\sin^2\theta) = 0 \Rightarrow \theta = \ldots$	**M1**	Sets their $\frac{\mathrm{d}y}{\mathrm{d}\theta}$ to zero and solves by trigonometric identities, reaching $\theta = \ldots$. Not for substituting $\theta = \frac{\pi}{6}$ to verify.
	$\sin\theta = \frac{1}{2} \Rightarrow \theta = \frac{\pi}{6}$ $r = 1 + 2\cos\frac{\pi}{3} = 2$	**A1 AG**	Correct equation solved without error to $\theta = \frac{\pi}{6}$ as the only solution in range, with $r = 2$ shown by substitution.
		[4]	
6(b)	$\frac{1}{2}\int_0^{\pi/6}(1 + 2\cos 2\theta)^2\,\mathrm{d}\theta$	**B1**	May be seen within a larger expression.
	$\frac{1}{2}\int(3 + 4\cos 2\theta + 2\cos 4\theta)\,\mathrm{d}\theta$ $= \frac{1}{2}\left[3\theta + 2\sin 2\theta + \frac{1}{2}\sin 4\theta\right]_0^{\pi/6}$	**M1**	Expands, replaces $\cos^2 2\theta$ by $\frac{1}{2}(1 \pm \cos 4\theta)$ and integrates to the form $p\theta + q\sin 2\theta + s\sin 4\theta$. Condone sign and coefficient slips.
	$= \frac{1}{2}\left(\frac{\pi}{2} + \sqrt{3} + \frac{\sqrt{3}}{4}\right) = \frac{\pi}{4} + \frac{5\sqrt{3}}{8}$	**dM1 A1**	**dM1** Substitutes $\frac{\pi}{6}$ and 0 into their integrated expression and subtracts; implied by a correct value.
	$P = (\sqrt{3}, 1)$ $R = \frac{1}{2}(1)(3 - \sqrt{3})$ $\quad - \left[\text{“}\frac{\pi}{4} + \frac{5\sqrt{3}}{8}\text{”} - \frac{1}{2}(3)(2)\sin\frac{\pi}{6}\right]$	**M1**	Combines their sector area, correctly signed, with areas of triangle OAP and the triangle A, P, $(3, 1)$, or an equivalent polygon. Triangle areas need a correct method; condone arithmetic slips.
	$R = 3 - \frac{\pi}{4} - \frac{9\sqrt{3}}{8}$	**A1**	Any exact equivalent.
		[6]	
	Total for Question 6	**10**	

Question	Scheme	Marks	Notes
7(a)	$x = r\cos\theta,\ y = r\sin\theta \Rightarrow \frac{\mathrm{d}x}{\mathrm{d}\theta} = \ldots,\ \frac{\mathrm{d}y}{\mathrm{d}\theta} = \ldots$	**M1**	Differentiates $r\cos\theta$ or $r\sin\theta$ by the product rule, treating r as a function of θ. Condone sign slips.
	$\frac{\mathrm{d}x}{\mathrm{d}\theta} = \frac{\mathrm{d}r}{\mathrm{d}\theta}\cos\theta - r\sin\theta$ $\frac{\mathrm{d}y}{\mathrm{d}\theta} = \frac{\mathrm{d}r}{\mathrm{d}\theta}\sin\theta + r\cos\theta$	**A1**	
	$\tan\psi = \frac{\mathrm{d}y}{\mathrm{d}x} = \frac{\mathrm{d}y/\mathrm{d}\theta}{\mathrm{d}x/\mathrm{d}\theta}$	**M1**	Uses $\tan\psi = \frac{\mathrm{d}y}{\mathrm{d}x}$ as their $\frac{\mathrm{d}y}{\mathrm{d}\theta}$ divided by their $\frac{\mathrm{d}x}{\mathrm{d}\theta}$.
	$\tan\psi = \frac{\frac{\mathrm{d}r}{\mathrm{d}\theta}\sin\theta + r\cos\theta}{\frac{\mathrm{d}r}{\mathrm{d}\theta}\cos\theta - r\sin\theta}$	**A1 AG**	Correct derivatives substituted, leading without error to the printed result; combined steps accepted.
		[4]	
7(b)	$\tan\psi = \frac{\tan\theta + \frac{r}{r'}}{1 - \tan\theta\frac{r}{r'}}$, where $r' = \frac{\mathrm{d}r}{\mathrm{d}\theta}$	**M1**	Rearranges into the form $\frac{\tan\theta + k}{1 - k\tan\theta}$, or otherwise uses the compound angle formula for tan to relate ψ, θ and $\frac{r}{\mathrm{d}r/\mathrm{d}\theta}$. Condone sign slips.
	$\tan\psi = \tan(\theta + \beta)$, $\tan\beta = \frac{r}{\mathrm{d}r/\mathrm{d}\theta}$	**A1**	oe, such as $\tan(\psi - \theta) = \frac{r}{\mathrm{d}r/\mathrm{d}\theta}$.
	$\beta = \alpha \Rightarrow \frac{\mathrm{d}r}{\mathrm{d}\theta} = r\cot\alpha$	**A1**	Identifies the angle between OP and the tangent with α. Any equivalent form, such as $\tan\alpha = \frac{r}{\mathrm{d}r/\mathrm{d}\theta}$.
	$\int \frac{1}{r}\,\mathrm{d}r = \int \cot\alpha\,\mathrm{d}\theta$	**M1**	Separates their variables and integrates at least one side correctly.
	$\ln r = (\cot\alpha)\theta + c$	**A1**	Must include a constant. oe, such as $r = Ke^{(\cot\alpha)\theta}$.
	$\theta = 0,\ r = 2 \Rightarrow c = \ldots\ [= \ln 2]$	**M1**	Substitutes $\theta = 0$, $r = 2$ into their integrated equation, which must contain a constant, to find the constant.
	$r = 2e^{(\cot\alpha)\theta}$	**A1 AG**	cso. Correct constant, leading without error to the printed equation.
		[7]	
Total for Question 7		**11**	

Question	Scheme	Marks	Notes
8(a)	$A(6,0)$	**B1**	
		[1]	
8(b)	$y = r\sin\theta = \frac{6\sin\theta}{2-\cos\theta}$ $\frac{\mathrm{d}y}{\mathrm{d}\theta} = \frac{6\cos\theta(2-\cos\theta)-6\sin^2\theta}{(2-\cos\theta)^2}$	**M1 A1**	**M1** Differentiates $\frac{6\sin\theta}{2-\cos\theta}$ by the quotient or product rule, or $r\sin\theta$ by the product rule. Condone sign slips. Not for differentiating r or $r\cos\theta$. **A1** Any correct form; need not be simplified.
	$12\cos\theta - 6(\cos^2\theta + \sin^2\theta) = 0$ $\Rightarrow \cos\theta = \ldots$	**M1**	Sets the numerator of their quotient-rule derivative to zero, uses $\sin^2\theta + \cos^2\theta = 1$ and solves for θ.
	$\theta = \frac{\pi}{3}$	**A1**	Or awrt 1.05, with no other value in $0 \leq \theta \leq \pi$.
	$r = \frac{6}{2-\cos\text{“}\frac{\pi}{3}\text{”}}$ $\Rightarrow B\left(4, \frac{\pi}{3}\right)$	**M1 A1**	**M1** Substitutes their θ from $\frac{\mathrm{d}y}{\mathrm{d}\theta} = 0$ into the equation of C. **A1** Or $(4,$ awrt $1.05)$.
		[6]	
8(c)	$2r - r\cos\theta = 6 \Rightarrow 2r = x + 6$ $\Rightarrow 4(x^2+y^2) = (x+6)^2$	**M1**	Uses $x = r\cos\theta$ and $r^2 = x^2 + y^2$ to reach an equation in x and y only. Condone slips in squaring.
	$3(x-2)^2 + 4y^2 = 48$	**A1**	oe, such as $3x^2 + 4y^2 - 12x - 36 = 0$ or y as a function of x.
	$\ell_1 : y = 2\sqrt{3}, \quad \ell_2 : x = 6$	**B1**	Both.
	Area $= \int_2^6 \left(2\sqrt{3} - \frac{\sqrt{3}}{2}\sqrt{16-(x-2)^2}\right)\mathrm{d}x$	**M1 A1**	**M1** Forms line minus curve from $x = 2$ to 6, or $8\sqrt{3}$ minus $\int_2^6 y\,\mathrm{d}x$, with y the positive root of their Cartesian equation. An equivalent integral in y accepted. **A1** Correct integrand and limits, oe.
	$x - 2 = 4\sin t, \ \mathrm{d}x = 4\cos t\,\mathrm{d}t$ $\Rightarrow \int_0^{\pi/2} 16\cos^2 t\,\mathrm{d}t$	**M1 A1**	**M1** Uses $x - 2 = 4\sin t$ or an equivalent trigonometric substitution, replacing $\mathrm{d}x$, to reach an integral in t only. Condone coefficient slips. **A1** Correct integrand in t, any constant factor carried correctly; limits not required.
	$8\int_0^{\pi/2}(1+\cos 2t)\,\mathrm{d}t$ $= 8\left[t + \frac{1}{2}\sin 2t\right]_0^{\pi/2}$	**M1**	Uses $\cos^2 t = \frac{1}{2}(1 \pm \cos 2t)$ and integrates to the form $at + b\sin 2t$. Condone sign and coefficient slips.

Question 8 continued

Question	Scheme	Marks	Notes
	$8\sqrt{3} - \frac{\sqrt{3}}{2} \times 8\left(\frac{\pi}{2}\right) = \ldots$	**dM1**	Applies limits in t matching their substitution, or 2 and 6 after reverting to x, and combines with their rectangle term.
	$\sqrt{3}(8 - 2\pi)$	**A1**	cso. $\sqrt{3}(8 - 2\pi)$, or $p = 8$ and $q = -2$ stated; $8\sqrt{3} - 2\pi\sqrt{3}$ left in that form scores A0.
		[10]	
	Total for Question 8	**17**	